Bogleheads

Tax-adjusted asset allocation

*This article contains details **specific to United States (US) investors**. It does not apply to non-US investors.*

If you have accounts with different tax treatment (taxable, traditional IRA or 401(k), Roth IRA or 401(k)), equal dollar amounts in those accounts have different after-tax values. Therefore, if you want to optimize the after-tax value of your portfolio, you should base your asset allocation on the after-tax value of the accounts.

Why to adjust

The purpose of asset allocation is to design a portfolio with the desired level of expected return and risk. Returns and mathematical definitions of risk should be based on after-tax values, since the after-tax value is what you will be able to spend. Emotional definitions of risk should be based on how you perceive the risk.

Returns

The goal of returns is to get the money you will spend to achieve your goal. If you have $100K in stock in a Roth and the stock market doubles in value, you have $100K more to spend in retirement. If you have $100K in stock in a traditional IRA, you have $100K more pre-tax but will lose some of the value to taxes upon withdrawal. If you have $100K in stock in a taxable account, the account will be worth less than $200K after the stock market doubles in value, because of taxes on dividends, and you will have even less to spend because you will pay tax on your capital gains.

Mathematical risk

Mathematical risk is based on your need and ability to take risk. Your need to take risk is determined by the returns you require; if you need your portfolio to double to provide an adequate standard of living, you actually need it to double in after-tax value. Your ability to take risk is determined by the consequences of losses; losing $100K in your Roth IRA will reduce your standard of living (or require more additional savings to keep the same standard) by more than losing $100K in your traditional IRA or taxable account does.

Emotional risk

Emotional risk is determined by how you will react to losses. If you pull out of the stock market after it falls, or you cannot sleep during a bear market, then your portfolio is too risky. In theory, the best way to handle emotional risk is the same as any other risk; if your portfolio software is set up to only

present the after-tax value of your portfolio, then a $100K loss in an account taxed at 24% is only a $76K decrease in your bottom line. In practice, most investors will not be able to make these adjustments, particularly because the investment statements show dollar values.

How much to adjust

There are two reasonable formulas for adjustment, based on the marginal and overall tax rates. The marginal rate is the correct rate to use for considering changes in the asset allocation (and thus for most tax-adjusted asset allocation), while the overall rate is correct for estimating the portfolio value or expected returns.

Adjusting by the marginal rate

You should value your Roth at its full value, and reduce the value of your traditional IRA or 401(k) by your expected tax bracket at retirement (including state taxes if your state taxes these withdrawals, which varies by state). You should also reduce the value of your taxable account by a smaller percent, to account for future capital gains and dividend taxes; estimated values are necessary because the tax cost depends on how much you will have in actual capital gains.

If all of your IRA or 401(k) withdrawals will be in a 24% marginal rate when you retire, then 24% of your withdrawals from a traditional IRA or 401(k) will be lost to taxes. Therefore, it makes sense to assume that the IRS owns 24% of the account; whatever the gain or loss in nominal dollars, 76% will go to you and 24% to the IRS. In contrast, the IRS owns none of your Roth, which is tax-free.

If your IRA or 401(k) is large enough that the withdrawals will span multiple tax brackets each year, the marginal tax rate is the rate which affects the risk and return of any marginal decisions you might make, such as a change in your asset allocation. For example, if you expect that half your withdrawals will be taxed at 12% and half at 22%, you will lose only 17% of the total to taxes. However, if you increase the risk of your IRA investments and earn an extra $1,000, you will pay an extra $220 in taxes, not $170; if the risk works out badly and you lose $1,000, you will pay $220 less in taxes. Thus, in this situation, the IRS owns 22% of the part of the IRA in which your decision will make a difference.

The IRS also owns part of your taxable account, representing the difference between what you earn on it and what you could have earned in a tax-free account. If your taxable account is invested in stocks, you will lose some money every year you hold it to the tax on the dividends, and more to the capital gains tax when you sell. If it is invested in municipal bonds, you effectively lose the difference between yields on municipal bonds and corporate bonds of comparable risk every year. The calculation is more complicated (and depends on assumptions about future returns and tax rates), but the adjustment usually comes out about the same for a taxable account and an IRA; this spreadsheet (https://docs.google.com/spreadsheets/d/1E28C1YTb1jzi3hSKK3ABGvWgTmjSlO5W/edit?usp=sharing&ouid=103809420707053067543&rtpof=true&sd=true) can make an estimated calculation. The spreadsheet calculates the marginal tax rate, for the same reason as above.

Adjusting for the overall tax rate

The overall tax rate, averaged over all your withdrawals, determines the actual after-tax value of your accounts, and thus affects the after-tax returns. A 401(k) or traditional IRA loses less to tax adjustment from the overall tax rate; a taxable account loses about the same amount because its taxes are at a mostly-flat rate on capital gains and qualified dividends.

A natural assumption for the overall tax rate is that you will have some taxable income independent of your retirement accounts (such as 85% of Social Security, which is taxable for higher-income retirees), and your IRA or 401(k) will be taxed on top of that. Estimate your other income and deductions, and then compute the tax on the difference between that income and the retirement accounts. This will be somewhat less than your marginal tax rate if your IRA or 401(k) spans multiple brackets.

If you have a taxable account, you will probably still need to adjust based on your marginal tax rate, because the tax on qualified dividends and long-term gains is effectively a flat 15% (plus state tax) in most tax brackets. If your total taxable income in retirement is in a 15% tax bracket, you will not owe any federal tax on qualified dividends and capital gains in retirement. If you hold munis, the tax cost is unaffected by your retirement tax bracket, since it is an effective cost equal to the difference between taxable and tax-exempt bonds of comparable risk.

Why to make these adjustments

It is easiest to see why the adjustment is necessary by comparing traditional and Roth accounts. For example, suppose you have $4,000 in a 401(k) and $3,000 in a Roth IRA, and you will be in a hypothetical (for simplicity) 25% marginal rate when you retire. You might choose to invest the entire 401(k) in stock and the entire Roth in bonds. If you do this, and the stock market gains 10% while the bond market gains 5%, your 401(k) would be worth $4,400, and your Roth would be worth $3,150. The IRS will take 25% of your 401(k) when you withdraw it, so you could now withdraw your investments for $6,450. Similarly, if you invested the 401(k) in bonds and the Roth in stock, your 401(k) would be worth $4,200, and your Roth would be worth $3,300. After taxes, you would have the same $6,450. Since both portfolios give you the same 7.5% return that you would expect from a portfolio which was 50% stock and 50% bonds, it is reasonable to treat both portfolios as having a 50/50 asset allocation, rather than the 57/43 and 43/57 which would result from the nominal values.

The tax adjustment also adjusts for the effective difference in risk between the two accounts. In the example above, suppose that the stock market loses half its value. If your 401(k) was entirely in stock, it would drop from $4,000 to $2,000, which would cost you $1,500 in after-tax value; if your Roth was entirely in stock, it would drop from $3,000 to $1,500. Either way, you lose 25% of the value of your portfolio, so your portfolio has the risk of a portfolio which is 50% stock.

Consequences

Asset location

Once you adjust for the after-tax value, it does not matter which assets you put in a traditional IRA or 401(k) and which you put in a Roth. If you are in a hypothetical 25% marginal tax rate range, investing $4,000 in the 401(k) or $3,000 in the Roth in the same investment will give you the same after-tax value.

If all else is equal (and it often isn't because of limited 401(k) options), it is slightly better to put assets with higher risk in the Roth, even after adjusting for after-tax value. The reason is that the progressive tax code creates asymmetrical risk; you can be in a situation in which the IRS takes a larger percentage of your traditional gains than it gives back of your losses. If you know you will be withdrawing at a 22% marginal tax rate, then $100,000 in traditional and $78,000 in Roth are equivalent; for every 1% that you gain or lose in either account, you have $780 more or less to spend. But if the traditional account is in stock, you may gain or lose enough multiples of $1,000 that your withdrawals move to a different tax bracket. If the stock market booms, some of the gains of $1,000 will be taxed at 24%, giving you only $760 more to spend; if it crashes, some of the losses of $1,000 will reduce the part of the withdrawal that would be taxed at 12%, costing you $880 in spending money.

In addition, the Roth is protected against potential changes in tax rates, has more flexible rules for required minimum distributions, and is not counted as income for making Social Security taxable.

It still does matter which assets you put in your taxable accounts; see Tax-efficient fund placement for details.

Roth conversions

You can evaluate a conversion of a traditional IRA to a Roth by its effect on your after-tax asset allocation. If you are in the same tax bracket now that you expect to be in at retirement, and you pay taxes on the conversion with IRA money (or with money you would have otherwise contributed to a 401(k) or IRA), the conversion is break-even. For example, if you are at a hypothetical 25% tax marginal rate and convert a $40,000 IRA, you will have $30,000 in the Roth after paying taxes. Previously, you owned $30,000 of the IRA and the IRS owned the other $10,000; after conversion, you own the entire $30,000 in the Roth.

In contrast, if you pay the taxes with taxable money, you have a net gain, and it may even be worth making the conversion if you are going to be in a slightly lower tax bracket at retirement. If you are in a hypothetical 28% marginal rate now but expect to retire at a 25% marginal rate, and have $11,200 in a taxable account and $40,000 in an IRA, you own 75% of the IRA and probably about 75% of the taxable account, a total of $38,400. If you convert to a Roth, paying the taxes with the taxable $11,200, you will have a Roth worth $40,000.

The advantage of a Roth

If you can max out either a traditional or a Roth IRA or 401(k), you effectively get more tax-advantaged money with the Roth. You can contribute the same number of dollars either way, but you own all of the Roth and not all of the traditional account. See "Traditional plus taxable" vs. Roth for details.

Investing in the Roth in this situation is equivalent to investing in a traditional IRA and immediately converting it to a Roth, paying the tax with after-tax dollars. Therefore, if you are in a much higher tax bracket now than you will be in at retirement, the "traditional plus taxable" accounts will be better,

but if the brackets will be equal or close, you should prefer the Roth. What constitutes "much higher" and "close" is situation-dependent, and the wiki section mentioned in the paragraph above suggests some tools to help evaluate that.

See also

- Tax-efficient fund placement

External links

- Asset allocation spreadsheet which adjusts for after-tax asset allocation (https://docs.google.com/spreadsheets/d/1E28C1YTb1jzi3hSKK3ABGvWgTmjSlO5W/edit?usp=sharing&ouid=103809420707053067543&rtpof=true&sd=true), Google Sheets
- Do Your Asset Classes Care Where They Are? (https://www.efficientfrontier.com/ef/704/where.htm), William Bernstein, *Efficient Frontier*
- Tax-Adjusted Asset Allocation (https://www.obliviousinvestor.com/tax-adjusted-asset-allocation/), Oblivious Investor blog.
- Exploring the Bogleheads(TM) Wiki - Tax-Adjusted Asset Allocation (https://www.forbes.com/sites/thebogleheadsview/2012/04/19/exploring-the-bogleheadstm-wiki-tax-adjusted-asset-allocation/), Mel Lindauer on Forbes.
- Bogleheads forum topic: "Roth vs Traditional asset location - advanced discussion" (https://www.bogleheads.org/forum/viewtopic.php?t=66960). January 18, 2011
- Bogleheads forum topic: "Grok Tip 15 You're doing it wrong: Figuring stock/bond split" (https://www.bogleheads.org/forum/viewtopic.php?t=124899)
- Bogleheads forum topic: "Problems with Reichenstein's "Asset Location Decis Revisted" " (https://www.bogleheads.org/forum/viewtopic.php?t=126195)

Bibliography

- James M. Poterba, "Saving for Retirement: Taxes Matter (https://ssrn.com/abstract=546662)", (May 2004). Boston College Center for Retirement Research Issue Brief No. 17. Available at SSRN.
- C. Reed, "Rethinking Asset Location - Between Tax-Deferred, Tax-Exempt and Taxable Accounts (https://ssrn.com/abstract=2317970)", (August 28, 2013). Available at SSRN.
- William E. Reichenstein, "Asset Allocation and Asset Location Decisions Revisited (https://www.efficientfrontier.com/ef/704/REICHENSTEINJWMMarkOpt.pdf)", Journal of Portfolio Management (Summer 2001).

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